

A car of proper length $L_0 = 5$ m travels at speed $v = 0.8c$ toward a garage of rest length 3 m. Bob is at rest with the garage.

The garage lies along the x -axis from $x = 0$ (back wall) to $x = 3$ m (front door). The car approaches the garage from the *right* (from positive x) and moves toward *decreasing* x (i.e. it moves left). Bob is standing at the back wall at $x = 0$. Alice is inside the moving car.

1. **(2 marks)** Show that, as measured by Bob, the car fits inside the garage.
2. **(3 marks)** Define event A as: the *front bumper* of the car reaches the back wall at $x = 0$. Define event B as: the *tail end of the car* reaches the front door at $x = 3$ m (the instant when the entire car has just entered the garage). Bob observes A and B to occur simultaneously.

Choose convenient coordinates for A and B in Bob's frame and determine how Alice orders these two events. Compute the time difference $\Delta t'$ between A and B as measured by Alice.

3. **(2 marks)** Using your result from part (b), explain why there is no contradiction: Bob can legitimately say the car fits inside the garage at one instant, while Alice does not see the car fully inside at any single instant in her frame.
4. **(3 marks)** When the front of the car reaches the back wall (event A), Bob can send a light signal from $x = 0$ toward the front door at $x = L_{\text{garage}}$. To trap the car the signal must reach the front door *before* the tail of the car reaches it (event B).
- (i) Working entirely in Bob's frame, determine the minimum garage length L_{min} such that a light signal sent at A can reach the front door before the tail arrives.
 - (ii) Derive a general expression for L_{min}/L_0 in terms of the speed v .
 - (iii) Briefly comment on how L_{min}/L_0 compares with the Lorentz factor γ .